Commends

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Part A - Downloading E. Coli WGS data, Preliminary Analyses, and Quality Controls:

1. Install SRA Toolkit (for downloading sequencing data)

   sudo apt-get update

   sudo apt-get install sra-toolkit

2. Install FastQC (for quality control):

   sudo apt-get install fastqc

3. Create a folder for project:

   mkdir Project1

   cd Project1

   mkdir input_data qc_results

4. Download the sequencing data:

   fastq-dump --split-files SRR8185317

5. Count the number of reads:

   grep -c "^@" SRR8185317_1.fastq

6. Print the first read:

   head -n 4 SRR8185317.fastq

7. Count occurrences of TTAAATGGAA:

   grep -o TTAAATGGAA SRR8185317_1.fastq | wc –l

8. Extract the first 1000 reads:

   head -n 4000 SRR8185317_1.fastq > first1000.fastq

9. Quality and read length analysis:

   Code of plots in part A.r file

10. Perform Quality Control:

    Split the FASTQ File: head -n 4000 SRR8185317_1.fastq > subset.fastq

    fastqc subset.fastq -o ../qc_results/

Part B - De Novo Genome Assembly:

1. Install SPAdes:

```
sudo apt-get install spades
```

2. Run SPAdes for Genome Assembly:

```
cd ~/Project1/input_data
spades.py -s SRR8185317.fastq -o ../de_novo_assembly --threads 4 --isolate
```

3. Verify SPAdes Output:

```
cd ~/Project1/de_novo_assembly
ls
```

4. Install Quast:

```
wget https://github.com/ablab/quast/releases/download/quast_5.2.0/quast-5.2.0.tar.gz
tar -xvzf quast-5.2.0.tar.gz
cd quast-5.2.0
sudo apt-get update
sudo apt-get install python3 python3-pip
pip3 install numpy matplotlib
python3 setup.py install
```

5. Run Quast to evaluate the quality of the assembly:

```
python3 ~/Project1/input_data/quast-5.2.0/quast.py
/home/asal/Project1/de_novo_assembly/contigs.fasta -o ../assembly_quality
```

6. Review Quast Results:

```
cd ~/Project1/assembly_quality
ls
```

Part C - Read Mapping:

1. Install BWA:

```
sudo apt-get install bwa
```

2. Install Samtools:

```
sudo apt-get install samtools
```

3. Download the Reference Genome:

```
cd ~/Project1

mkdir reference_data

wget
ftp://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/005/845/GCF_000005845.2_ASM584v2/GCF_000005845.2_ASM584v2_genomic.fna.gz

gunzip GCF_000005845.2_ASM584v2_genomic.fna.gz

mv GCF_000005845.2_ASM584v2_genomic.fna reference.fasta
```

4. Index the Reference Genome:

```
cd ~/Project1/reference_data

bwa index reference.fasta
```

Mapping raw sequencing reads to a reference genome:

5. Map Reads to the Reference Genome:

```
mkdir -p ~/Project1/mapping_results

cd ~/Project1/input_data

bwa mem ~/Project1/reference_data/reference.fasta SRR8185317.fastq >
../mapping_results/mapped_reads.sam
```

6. Obtain the Head of the SAM File:

```
head mapped_reads.sam
```

7. Convert SAM to BAM Format:

cd ~/Project1/mapping_results

samtools view -bS mapped_reads.sam > mapped_reads.bam

samtools sort mapped_reads.bam -o mapped_reads_sorted.bam

samtools index mapped_reads_sorted.bam

8. Visualize Mapped Reads by IGV:

Download, Instalation & Run: https://igv.org/doc/desktop/#DownloadPage/

tar -xvzf igv_<version>.tar.gz

cd IGV_Linux_2.19.1/igv.sh

./igv.sh

Load Files into IGV: Genomes > Load Genome from File > reference.fasta

File > Load from File > mapped_reads.bam

9. Evaluate Mapping Statistics:

samtools flagstat mapped_reads_sorted.bam

samtools depth mapped_reads_sorted.bam > read_depth.txt

awk '{sum += $3} END {print "Average Read Depth = ", sum/NR}' read_depth.txt

10. Compute Read Depth:

samtools depth mapped_reads_sorted.bam | awk '{sum+=$3} END {print "Mean read depth: ", sum/NR}'

Alignment Results of the Assembled Genome with the Reference Genome:

11. Align the Assembled Genome (Contigs) to the Reference Genome:

mkdir -p ~/Project1/ alignment_results_assembled_genome

cd ~/Project1/input_data

bwa mem reference.fasta contigs.fasta > aligned.sam

12. Obtain the Head of the SAM File:

```
head aligned.sam
```

13. Convert SAM to BAM Format:

```
cd ~/Project1/ alignment_results_assembled_genome
samtools view -bS aligned.sam > aligned.bam
samtools sort aligned.bam -o aligned_sorted.bam
samtools index aligned_sorted.bam
```

14. Visualize Mapped Reads by IGV:

Load Files into IGV: Genomes > Load Genome from File > reference.fasta

File > Load from File > aligned_sorted.bam

15. Evaluate Mapping Statistics:

```
samtools flagstat aligned_sorted.bam
samtools depth aligned_sorted.bam > read_depth.txt
```

16. Compute Read Depth:

```
samtools depth aligned_sorted.bam | awk '{sum+=$3} END {print "Mean read depth: ", sum/NR}'
```